Meridian Health Group — Patient Experience Team: Technology Context

Services Owned by This Team

Service	Responsibility	Technology
Appointment Service	Slot availability, booking, cancellation, rescheduling, waitlist, Epic sync	Lambda + PostgreSQL 15 + FHIR Adapter
Prescription Service	Repeat prescription request submission, status polling from Epic, reminder scheduling	Lambda + DynamoDB
Messaging Service	Patient-to-clinician async messaging, attachment handling, urgent escalation flag	Lambda + PostgreSQL 15 + S3
Telehealth Service	Doxy.me session lifecycle, waiting room state, recording consent	Lambda + DynamoDB
Payment Service	Stripe co-payment initiation, receipt, booking-payment atomicity, refund on Epic booking failure	Lambda + Stripe API

Frontends Owned by This Team

Client	Technology	Notes
Patient Web Portal	React 18 (PWA, Workbox)	Offline read-only cache for upcoming appointments and latest labs; WCAG 2.1 AA; axe-core in CI
Patient Mobile App	Capacitor shell wrapping the PWA	iOS + Android native wrapper; push notifications via FCM/APNs; no separate native codebase

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Key Technical Constraints

Appointment Booking + Co-Payment Atomicity

- Booking workflow is a two-phase transaction: (1) create Stripe PaymentIntent and collect payment, (2) write `Appointment` to Epic via FHIR Adapter
- If the FHIR write-back to Epic fails after payment is collected, the Appointment Service triggers a Stripe refund automatically (refund within 5 seconds of FHIR failure)
- If Stripe payment fails, the FHIR write is never attempted; booking is not confirmed
- Appointment slot is held with a **5-minute reservation lock** (Redis TTL) during payment collection to prevent double-booking

Lab Result Display

- FHIR Adapter polls Epic every 15 minutes for new `DiagnosticReport` and `Observation` resources
- Lab results are only surfaced to the patient **after Epic sets verification status** (`DiagnosticReport.status = final`)
- Results cached in Redis (15-minute TTL); cache invalidated on new poll results

- Patient receives a push + email notification when new lab results are available (Notification Service)

Proxy Access Enforcement

- Proxy grants stored in Cognito (custom attribute: `proxyFor: [patientId1, patientId2]`)
- Admin approval flow: proxy request stored in a DynamoDB queue; admin reviews via Care Coordination portal; approval triggers Cognito attribute update
- All API calls made with a proxy JWT include both `proxyUserId` and `patientId`; services validate that a valid proxy grant exists before returning PHI

Virus Scanning on Attachments

- Message attachments uploaded via pre-signed S3 URL (max 5 MB photos, 10 MB PDFs)
- S3 `ObjectCreated` event triggers a ClamAV Lambda; if a virus is detected, the object is deleted and the upload is rejected with a 400 response to the patient; Messaging Service notifies the patient

Telehealth (Doxy.me)

- Doxy.me session token generated server-side by Telehealth Service; valid for 60 minutes from appointment start time
- Doxy.me iframe embedded in the patient portal at appointment time; postMessage API used for session lifecycle events (joined, ended, recording-consent)
- Recording: opt-in consent captured before session starts; recordings stored in S3 with 90-day retention

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External Integrations (this team)

Integration	Protocol	Direction	Notes
FHIR Adapter → Epic	FHIR R4 REST (OAuth 2.0 SMART on FHIR)	Appointment/Prescription Service → FHIR Adapter → Epic	Booking, cancellation, prescription write-back; lab and radiology polling
Doxy.me	iframe + postMessage	Telehealth Service → Doxy.me	Session token exchange; lifecycle events
Stripe	Stripe Payment Intent API	Payment Service ↔ Stripe	Co-payment collection; refund on FHIR failure; webhook: `payment_intent.succeeded`, `payment_intent.payment_failed`
Twilio	REST API	Notification Service → Twilio	Appointment reminders (24h + 2h); urgent message alerts
SendGrid	REST API	Notification Service → SendGrid	Booking confirmations, lab available notifications, password reset
AWS Cognito	SDK	Auth Service ↔ Cognito	Patient registration, MFA, proxy attribute management
ClamAV	Lambda (event-driven)	S3 ObjectCreated → ClamAV Lambda	Virus scan all message attachments before delivery

Non-Functional Requirements (this team)

Category	Requirement
Appointment booking	Slot reservation lock held for 5 minutes during payment; FHIR write-back within 3 seconds of payment confirmation
Lab result latency	Results polled every 15 minutes; patient notified within 5 minutes of Epic verification
Attachment virus scan	ClamAV scan completes within 10 seconds; infected files deleted and patient notified before delivery
Co-payment refund	Stripe refund initiated within 5 seconds of FHIR write-back failure
Telehealth session	Doxy.me session token generated within 2 seconds of patient joining the waiting room
Offline PWA	Upcoming appointments (next 7 days) and latest lab results cached for offline read access
Accessibility	WCAG 2.1 AA; axe-core integrated in GitHub Actions CI; screen reader tested on iOS VoiceOver and Android TalkBack
HIPAA audit	Every PHI access event emitted to Audit Service within 1 second (EventBridge async)